

Within Marriage Age Gap Across Countries

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Abstract. The age gap within marriage is one form of the gender gap. This paper uses micro data from 89 countries from all income levels to document that men marry women younger than themselves in all countries, and the age gap decreases with development. We also show that the declining pattern is robust for both females and males after controlling for individual- and country-level characteristics.

JEL Classification: J11, J12, O11

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1 Introduction

Dramatic improvements in women’s economic and political rights come along with economic development in almost all industrialized countries (Doepke and Tertilt, 2016). It is well known that gender gaps in education, wage, and political rights decrease with development. By drawing on a new micro-level dataset covering countries from all income levels, we focus on a new dimension of the gender gap in this paper: the age gap within a marriage, defined as the husband’s age minus the wife’s age.

Using a new data set covering 204 nationally representative surveys from 89 countries from all income levels, we document that the age gap between the husband and the wife within a marriage decreases with GDP per capita across countries. Specifically, one log point increase in GDP per capita is associated with an average decline of 1.6 years in the husband-wife age gap in a fitted linear trend. To interpret this magnitude, consider, for example, Liberia, whose average GDP per capita is four log points lower than that of the U.S. in our sample. The average age gap of 8 years in Liberia is predicted to be 6.4 years higher than that in the U.S. by the linear trend, which is consistent with the 6 years’ difference in age gaps in our sample. We also find that, over the life cycle, the marriage age gaps vary drastically with the age of females and males in low- and middle-income countries, ranging from -15 to 20 years, while they remain relatively stable between -5 and 7 years in high-income countries.

We further combine 81 million females and 80 million males in our sample to test the new pattern after controlling for individual-level characteristics including age, education, rural-urban status, and polygynous status, as well as the relative supply of younger females to older males and the region fixed effects at the country-level. The individual-level regressions show that the negative and significant coefficient of marriage age gaps on log GDP per capita survives, ranging between -1.8 and -0.6 when we sequentially add the control variables.

This paper relates to the literature on marriage patterns and economic development. Importantly, Moro et al. (2017) show that a country’s macroeconomic conditions play a key role in individuals’ family formation decisions. This paper also relates to studies on the marriage age gap, most of which use the aggregate data to measure the marriage age gap as the difference between the mean or median age of husbands and wives at marriage (see, e.g., Rios-Rull et al., 2016; Tertilt, 2005; Carmichael, 2011). One exception is Shephard (2019), who combines micro data in the U.S. and a quantitative framework to emphasize that the marriage age gap and behavior within the household are intimately linked. However, all the papers above only focus on a specific set of countries. Our paper complements the literature by utilizing household-level data for countries from all income levels to document the decline in the within-marriage age gap with development.

2 Data and Measurement

The *marriage age gap* is defined as the husband’s age minus the wife’s for couples who are currently married or in consensual unions. To measure it, we draw on the micro-level data from the *Integrated Public Use Microdata Series, International* (IPUMS-International). This data set includes harmonized censuses and surveys that cover age, gender, marital status, educational attainment, urban-rural status, and other individual-level characteristics. Moreover, it constructs two family interrelationship variables that link the husband and the wife who live in the same household: i) *pernum* as the person’s identification number in a household; ii) *sploc* as the identification number of one’s spouse. These two variables allow us to compute the husband-wife age gap as well as identify polygynous unions, which are the “marriages” between one husband and more than one wife. For males in polygynous unions, we take the average age of the wives to calculate the age gap. Finally, we measure the aggregate marriage age gap as the average of individual-level marriage age gaps weighted by the wife’s person weight. For *GDP per capita*, we use the output-side real GDP at chained PPPs in 2011 US\$ (rgdpo) from the *Penn World Table version 9.1* (PWT 9.1). The country-average measures are the average across all available years for one country.

We restrict our sample to data after 1990 to focus on cross-country comparisons rather than time-series fluctuations. Our final sample includes 204 country-year censuses and surveys across 89 countries from all income levels. Among them, 154 country-year censuses and surveys have variables about demographic characteristics and thus allow for individual-level regressions.

3 Results

We find that the marriage age gap is positive universally and is negatively associated with GDP per capita. Over the life cycle from age 15 to 89, the marriage age gap for females is (weakly) decreasing while that for males (weakly) increasing, and this gender disparity is more pronounced in low-income countries. Moreover, pooling 81 million females and 80 million males together, our individual-level regressions show that the development matters for the marriage age gap even after controlling for individual- and country-level characteristics. In particular, urbanization and monogamy depress the marriage age gap; older and more educated females tend to marry males with smaller age gaps.

Figure 1 plots the aggregate marriage age gap for all country-year observations and country averages against log GDP per capita. We restrict the sample to individuals who are at least 22 years old, which is the maximum of the minimum legally marriageable age worldwide

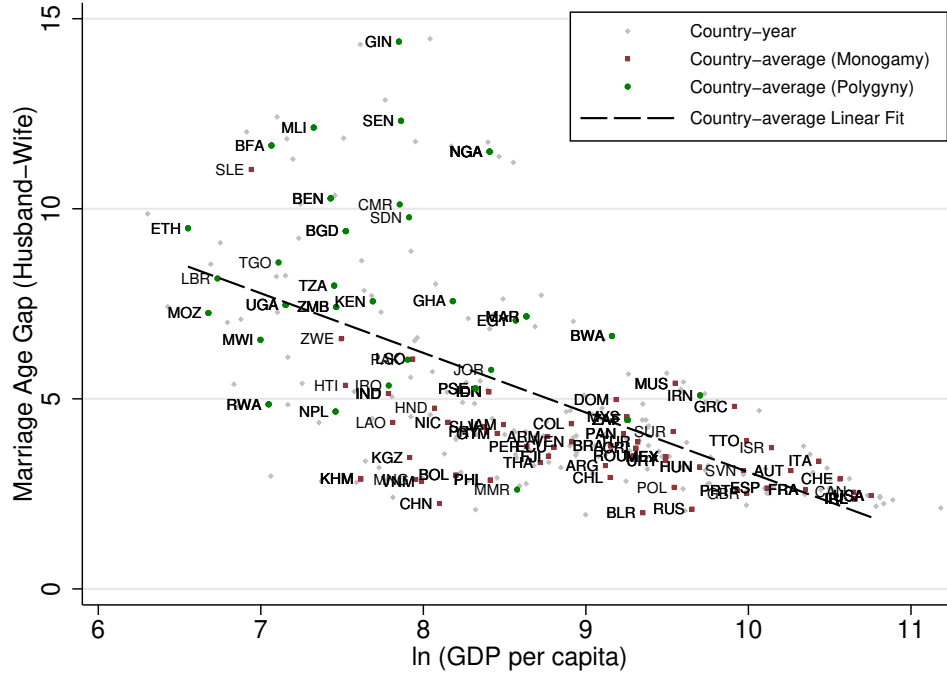


Fig. 1. Marriage age gap across countries.

according to the United Nations Statistics Division. Regressing the country-average marriage age gaps on log GDP per capita and a constant yields a slope coefficient of -1.6, which is statistical significance at the 1-percent level. In particular, the average marriage age gap is always positive in all of our 89 sampled countries. Moreover, as it decreases with economic development, it seems to stabilize at a gap of two or three years in the richest economies.

Figure 1 also shows that dispersion of the marriage age gaps across countries is larger at lower income levels, partially due to higher average age gaps in poor economies with polygynous unions. Adding a control variable of *polygyny dummy* decreases the coefficient of marriage age gaps on log GDP per capita to a significantly negative value at -0.9. In particular, for the most obvious seven outliers (Guinea, Burkina Faso, Mali, Senegal, Benin, Nigeria, and Cameroon), 37% of the married females and 17% of the married males are in polygynous unions. These ratios are much higher than the averages of 4.5% and 1.9% for married females and males, respectively, in the rest 21 countries where polygyny exists. Excluding these seven countries in the regression of country-average marriage age gaps on log GDP per capita yields a significant slope coefficient of -1.3.

We then compare the marriage age gaps over the life cycle by gender and income. Figure 2 plots the average marriage age gap of 5-year age bins from 15 to 89 years old for females and

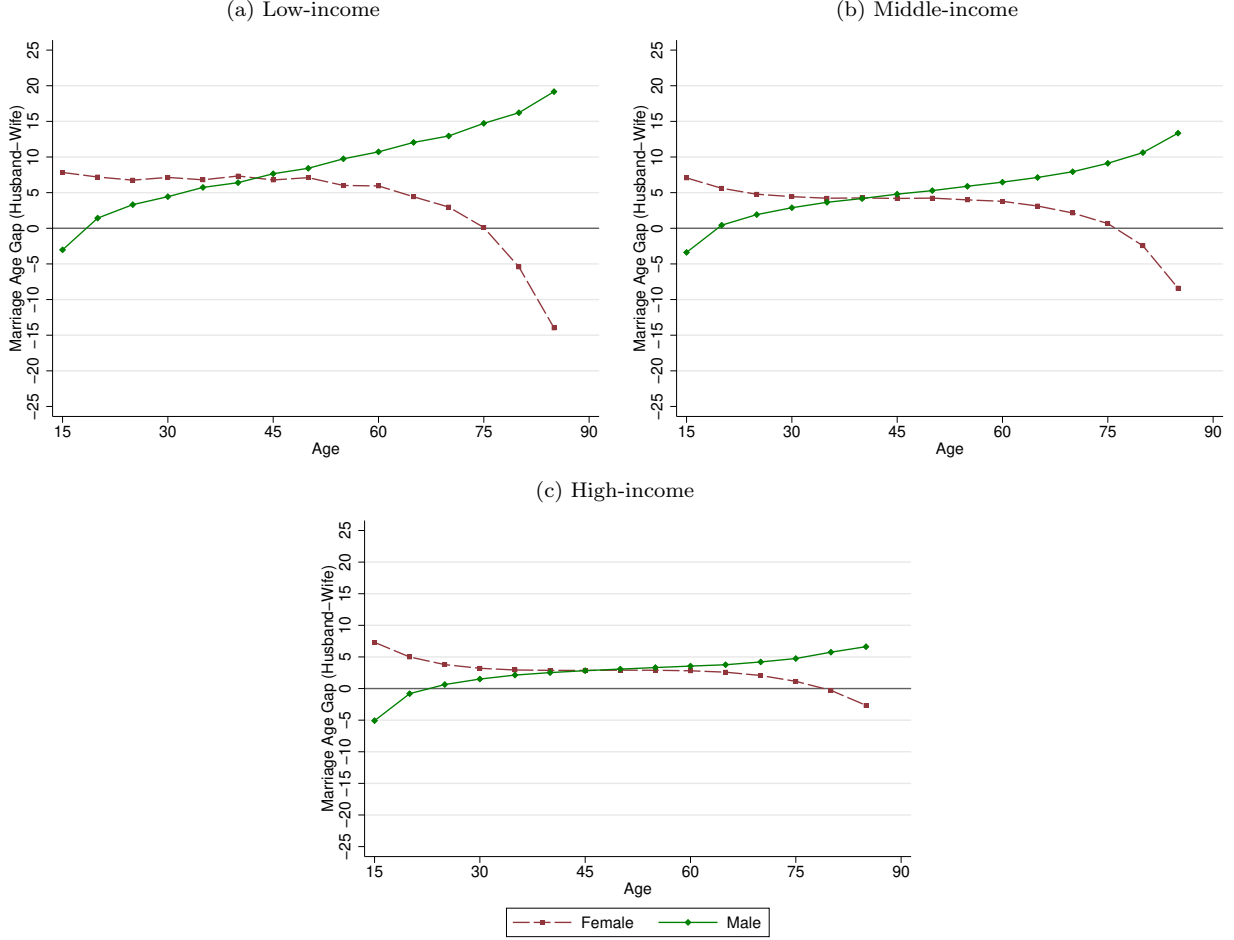


Fig. 2. Marriage age gap over life cycle by income.

males by income levels.¹ For all three income levels, the females' marriage age gap decreases with age while the males' increase, which results in the “X” shape of two lines in each panel. In general, young females aged 15-19 tend to marry males who are about 8 years older than them. Such age gap gradually decreases with age while keeping positive until age 79; above age 79, the age gap turns mostly negative and declines more drastically than before. This pattern is consistent with the fact that females have a longer life expectancy and a higher widowed rate than males (Feng and Ren, 2021). In contrast, only young males less than 20 years old tend to marry females older than themselves while older males are married to females younger than themselves, and their husband-wife age gaps increase almost linearly with age.

Figure 2 also highlights the differences in the life-cycle marriage age gap across income levels,

¹We divide the world income distribution into three terciles based on country-average GDP per capita since 1990 of all countries in the PWT 9.1. The thresholds of low-, middle-, and high-income groups are \$5,500 and \$15,200.

especially in the latter half of the life cycle. High-income countries have the mildest age gaps over the life cycle as indicated by the narrowest “X” pattern in the range between -5 and 7. It’s worth noting that females and males aged 35 to 64 in high-income countries have a stable husband-wife age gap at about 3 years. However, in low- and middle-income countries, the average marriage age gaps vary substantially over the life cycle.

To further investigate the relationship between the marriage age gap and development, we pool together over 160 million currently married individuals aged 22 and above. We then regress the husband-wife age gap on log GDP per capita and a constant for both genders, with control variables including age, educational attainment, the urban-rural status, a dummy of the polygynous union at the individual level, as well as a measure of the relative supply of younger females to older males and the the sub-region fixed effects at the country-level. The results are reported in Table 1, Panel A for females and Panel B for males.²

Table 1 shows that the coefficients of individual spousal age gaps on log GDP per capita range between -1.8 and -0.6, consistent with the magnitudes at the aggregate level. Moreover, the individual-level controls have reasonable effects on individuals’ marriage age gaps. Specifically, being one year older increases the marriage age gap of males by 0.16 years but decreases that of females by 0.05 years. In addition, higher educational attainment decreases females’ husband-wife age gap by 0.7 to 2.0 years compared to those who did not complete primary school, while the effects are much smaller for males with an indefinite sign. Meanwhile, living in urban areas decreases the age gap for both females and males, and the effect is more pronounced for males. Lastly, being in a polygynous union increases the age gap for females by 6.4 to 7.5 years compared to being in a monogamous marriage; the impacts on males are slightly smaller, lying between 5.1 to 6.0 years. Overall, adding individual-level characteristics decreases the absolute value of the coefficient on income levels by 42% for females, but increases that for males by 19%.

For country-level characteristics, we consider two variables in Columns 4 and 5 of Table 1. Firstly, to test the effect of the relative supply of younger females to older males, we add a control variable defined as the population ratio of females aged between 20 and 39 to males aged between 40 and 59 in each country-year survey. Note that this control variable positively correlates with population growth and also reflects sex ratios. We find that even increasing the younger females’ population by 100 percent of the older males’ population is associated with a very modest increase in the marriage age gap among males by 0.8 to 2 years, while the effect has indefinite signs for females. These results are consistent with the predictions in Bhaskar (2011, 2019) that age gaps are insensitive to imbalances of cohort sizes

²Countries in our data set locate in 13 out of 17 sub-regions according to the classification by United Nations Statistical Division.

Table 1. Individual-level Regression of *Marriage Age Gap*

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Female</i>					
ln (GDP per capita)	-1.56***	-1.01***	-0.91***	-0.74***	-0.58***
Age		-0.06***	-0.06***	-0.05***	-0.05***
Education					
Primary Completed		-1.38***	-1.25***	-1.19***	-0.70***
Secondary Completed		-1.74***	-1.65***	-1.55***	-1.41***
University Completed		-2.00***	-1.94***	-1.81***	-1.86***
Urban Dummy		-0.10***	-0.07***	-0.12***	-0.09***
Polygyny Dummy			7.45***	7.35***	6.35***
Pop. Ratio $\frac{\text{Younger Females}}{\text{Older Males}}$				0.69***	-0.230***
Sub-region Fixed Effects					✓
R^2	0.05	0.07	0.09	0.09	0.12
<i>Panel B. Male</i>					
ln (GDP per capita)	-1.48***	-1.80***	-1.76***	-1.26***	-0.92***
Age		0.15***	0.15***	0.16***	0.16***
Education					
Primary Completed		-0.56***	-0.52***	-0.34***	0.07***
Secondary Completed		-0.23***	-0.20***	0.12***	0.39***
University Completed		-0.02***	-0.003	0.31***	0.43***
Urban Dummy		-0.13***	-0.11***	-0.26***	-0.27***
Polygyny Dummy			5.92***	5.63***	5.09***
Pop. Ratio $\frac{\text{Younger Females}}{\text{Older Males}}$				1.98***	0.79***
Sub-region Fixed Effects					✓
R^2	0.05	0.17	0.18	0.19	0.21

Note: The data cover 80,629,271 females and 80,019,052 males. *** indicates statistical significance at the 1-percent level.

in the marriage market. Secondly, we control for sub-region fixed effects. The coefficients on sub-region dummies are the largest in Northern Africa, Sub-Saharan Africa, and Southern Asia, and are the smallest in Eastern Asia, Southern Europe, Central Asia, and Eastern Europe.

Overall, after controlling for the individual- and country-level characteristics, doubling GDP per capita is associated with declines of 0.6 and 0.9 years in the marriage age gaps for females and males, respectively.

4 Conclusions

Existing literature proposes various mechanisms to explain the observed marriage age gap such as gender gaps in career opportunities and fecundity (see, e.g., [Bergstrom and Bagnoli, 1993](#); [Díaz-Giménez and Giolito, 2013](#); [Coles and Francesconi, 2011](#)). This paper combines a large set of micro data to document that the age gap within marriage decreases with development and stabilizes at an age gap of two or three years. These new stylized facts need to be explained by marriage theories in future research.

References

- BERGSTROM, T. C. AND M. BAGNOLI (1993): “Courtship as a waiting game,” *Journal of political economy*, 101, 185–202.
- BHASKAR, V. (2011): “Sex selection and gender balance,” *American Economic Journal: Microeconomics*, 3, 214–44.
- (2019): “The Demographic Transition and the Position of Women: a Marriage Market Perspective,” *The Economic Journal*, 129, 2990–3024.
- CARMICHAEL, S. (2011): “Marriage and Power: Age at First Marriage and Spousal Age Gap in Lesser Developed Countries,” *The History of the Family*, 16, 416–436.
- COLES, M. G. AND M. FRANCESCONI (2011): “On the emergence of toyboys: The timing of marriage with aging and uncertain careers,” *International Economic Review*, 52, 825–853.
- DÍAZ-GIMÉNEZ, J. AND E. GIOLITO (2013): “Accounting for the timing of first marriage,” *International Economic Review*, 54, 135–158.

- DOEPKE, M. AND M. TERTILT (2016): “Families in macroeconomics,” in *Handbook of macroeconomics*, Elsevier, vol. 2, 1789–1891.
- FEENSTRA, R. C., R. INKLAAR, AND M. P. TIMMER (2015): “The Next Generation of the Penn World Table,” *American Economic Review*, 105, 3150–3182.
- FENG, Y. AND J. REN (2021): “Marriage and Development: Cross-country Evidence,” Working Paper.
- MINNESOTA POPULATION CENTER (2019): *Integrated Public Use Microdata Series, International: Version 7.2 [dataset]*, Minneapolis, MN: IPUMS, 2019.
- MORO, A., S. MOSLEHI, AND S. TANAKA (2017): “Marriage and economic development in the twentieth century,” *Journal of Demographic Economics*, 83, 379–420.
- RIOS-RULL, J.-V., S. SEITZ, AND S. TANAKA (2016): “Sex Ratios and Long-Term Marriage Trends,” Working Paper.
- SHEPHARD, A. (2019): “Marriage Market Dynamics, Gender, and the Age Gap,” Working Paper.
- TERTILT, M. (2005): “Polygyny, fertility, and savings,” *Journal of Political Economy*, 113, 1341–1371.